

Deep Learning for TikTok Video Popularity Prediction: An Variational Inference Perspective

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Joint work with Yongchun Li (UTK)

1. Introduction

Problem Setup (Feature Part)

1. Feature from Panel Data

	Publisher ID	Publisher Follower Count	Publisher Following Count	Publisher Total Heart Count	Publisher Total Video Count	Video Create Time	Video Description
1	A10	1.4M	6	7.2M	285	Jan-21-2022	A little throwback to the Mille
2	A11	22.3M	413	363M	1826	Feb-15-2022	#h2o #mermaid #netflix

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2. Feature from Visual

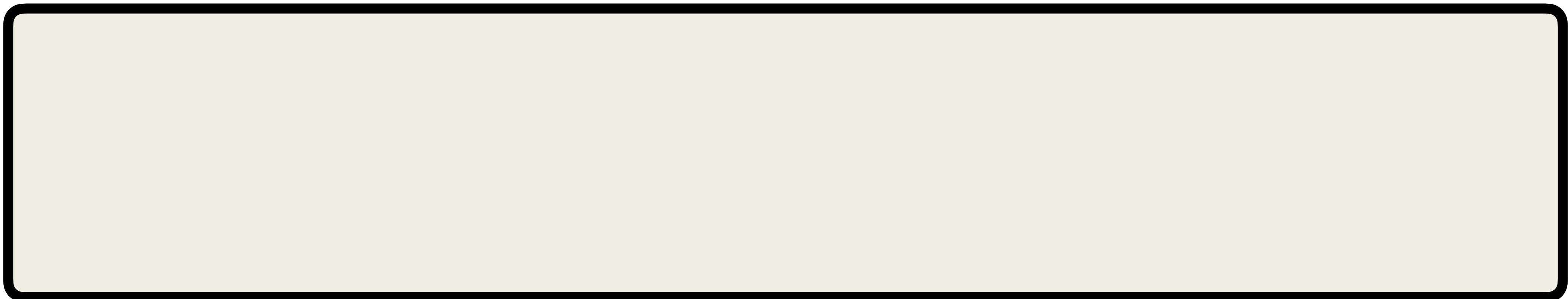


Problem Setup (Label Part)

	Video Comment Count	Video Heart Count	Video Play Count	Video Share Count
1	26	1241	15400	79
2	1331	84600	691600	1863

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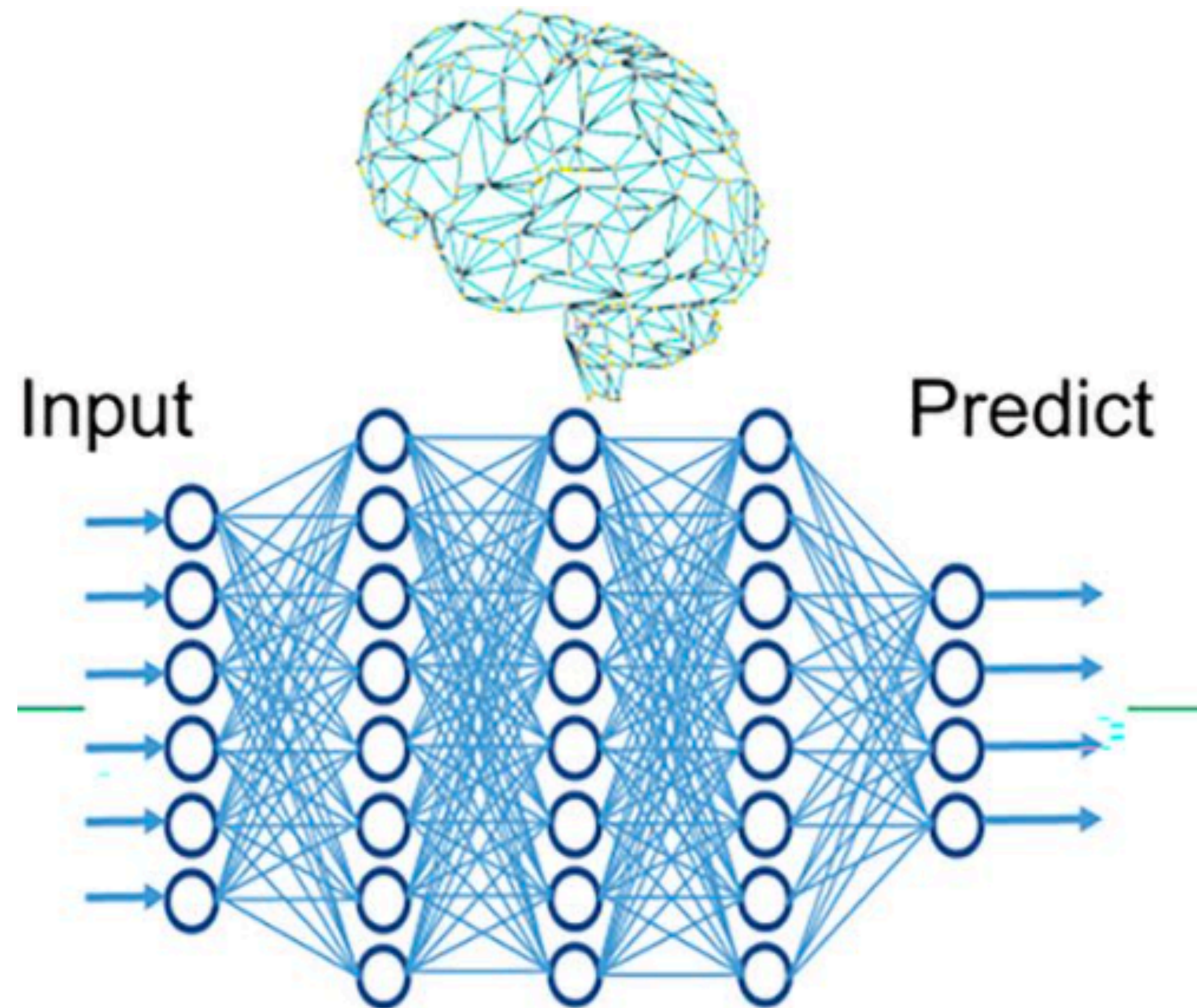
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Objective: Given training dataset consisting of features and labels, **Predict** the label of each feature from the testing dataset.

Classical Deep Learning Viewpoint

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n \left(f_{\text{NN}}(x_i; \theta) - y_i \right)^2$$

- Tabular
- Visual
- Acoustic
- Textual



- # of Comment
- # of Heart
- # of Play
- # of Share

Classical Deep Learning Viewpoint



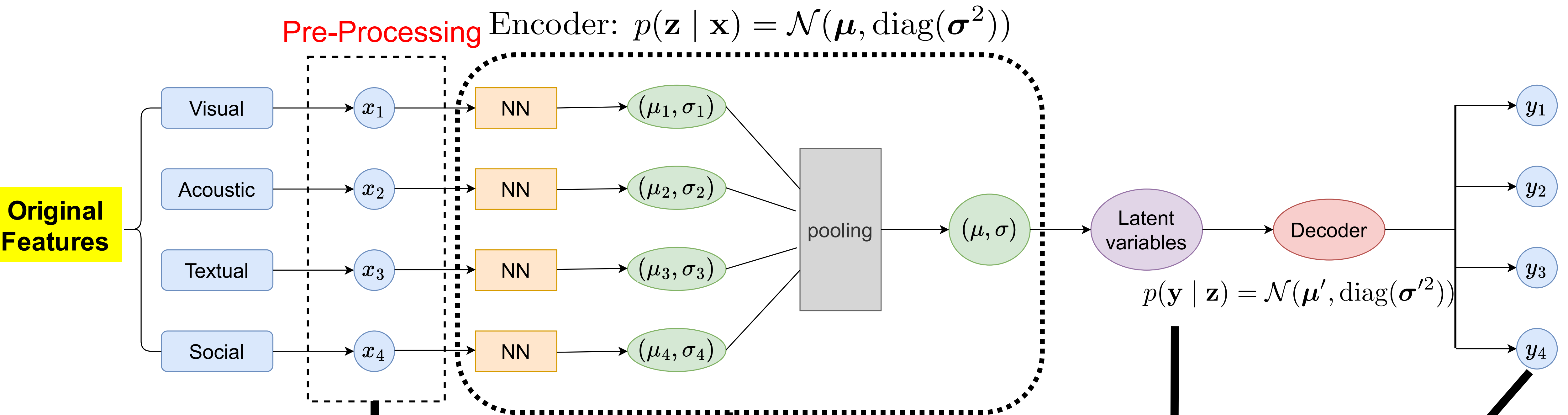
$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n \left(f_{\text{NN}}(x_i; \theta) - y_i \right)^2$$

Cons:

- Tabular
- Visual
- Acoustic
- Textual

- 1. Small-sample (2200 samples)**
- 2. High-dimensional (1280*720*100 pixels)**
- 3. Mixed features**
- 4. Black-box model, less of interpretability**

Our Approach



- **Pre-processing** of features
- **Feature**→**Latent Variable**→**Label** ($x \rightarrow y \rightarrow z$)
- Probabilistic representation for **uncertainty quantification**

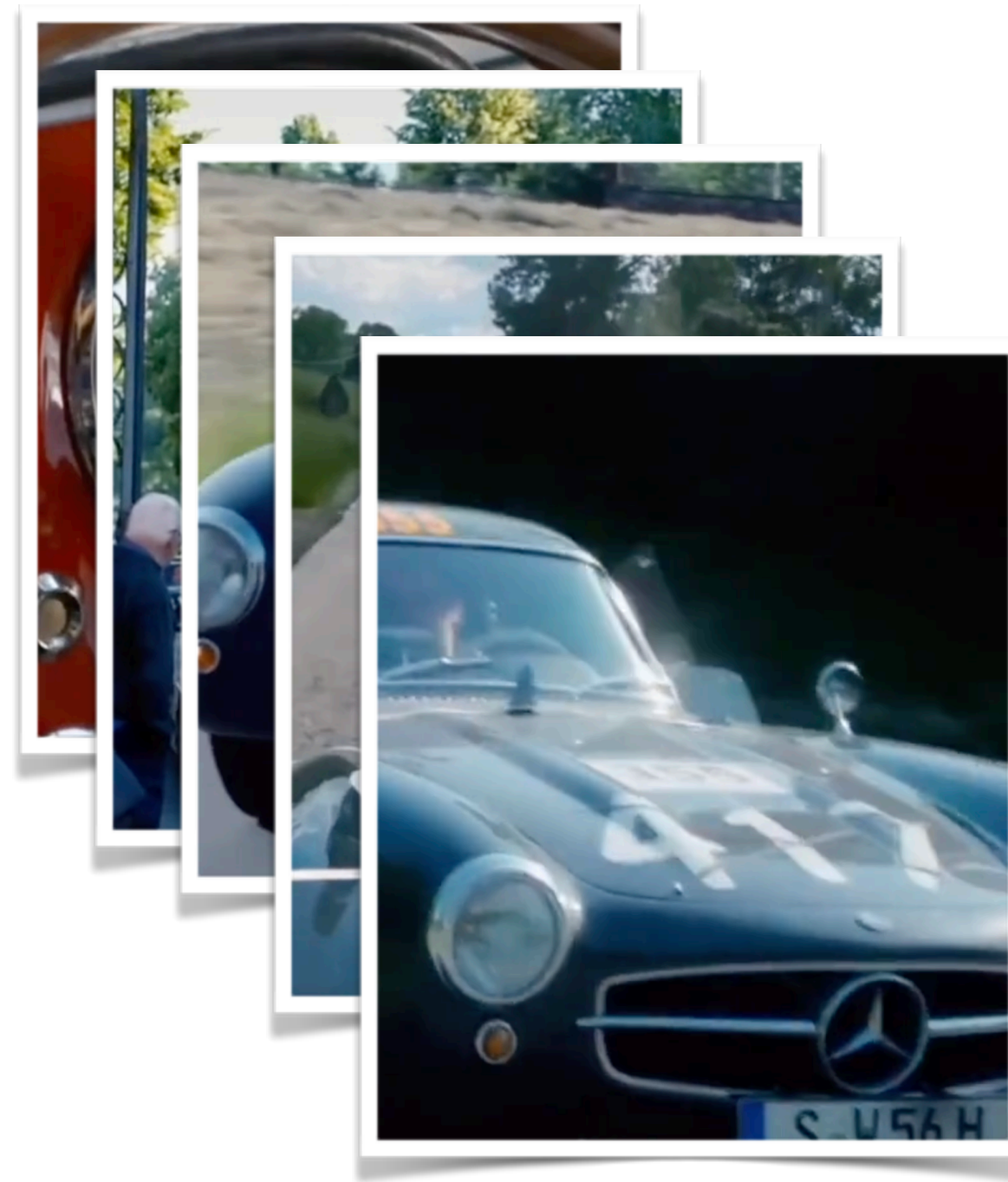
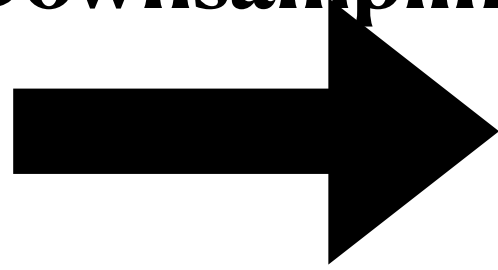
2. Pre-Processing

Visual Content

Visual Content



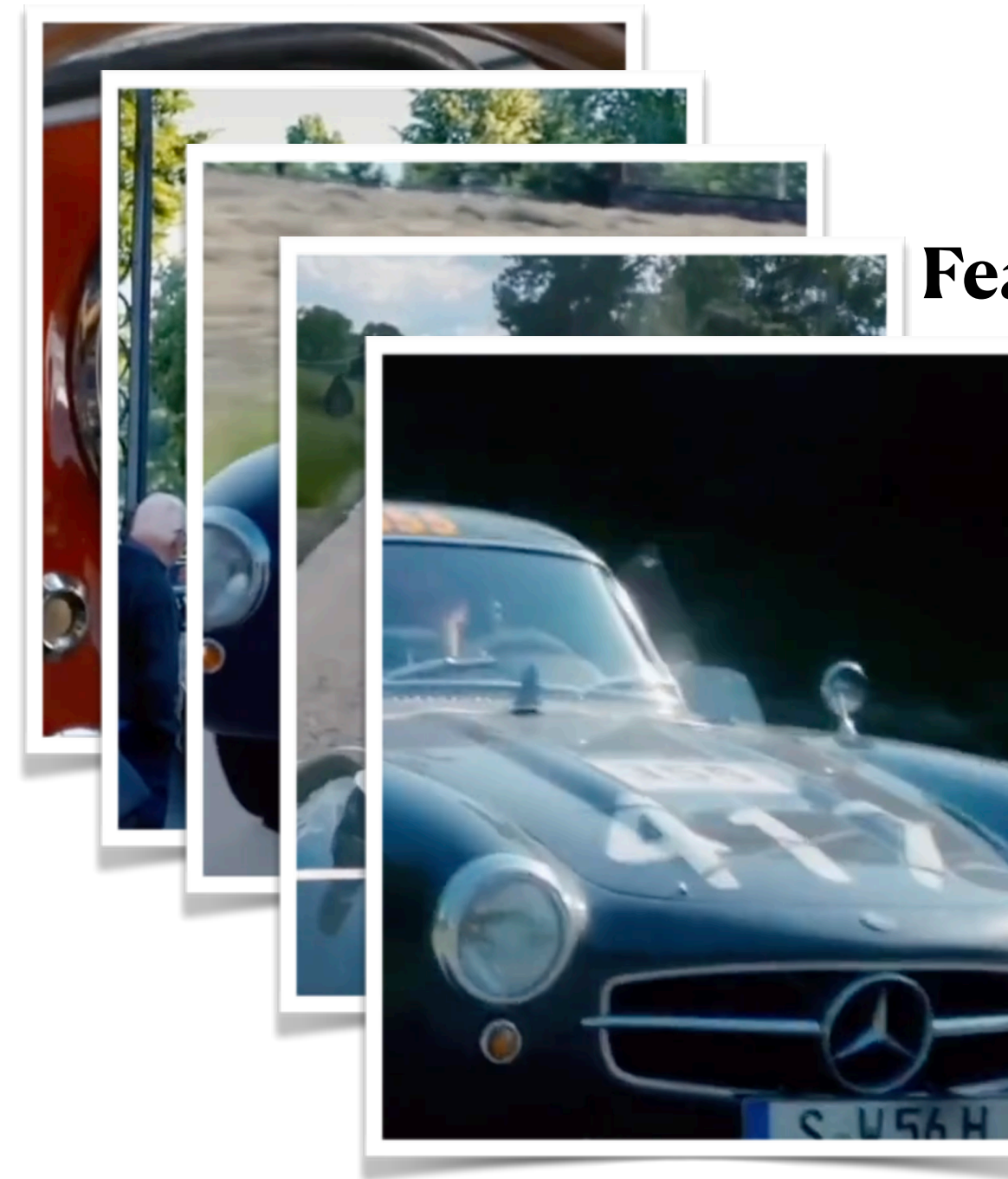
Downsampling



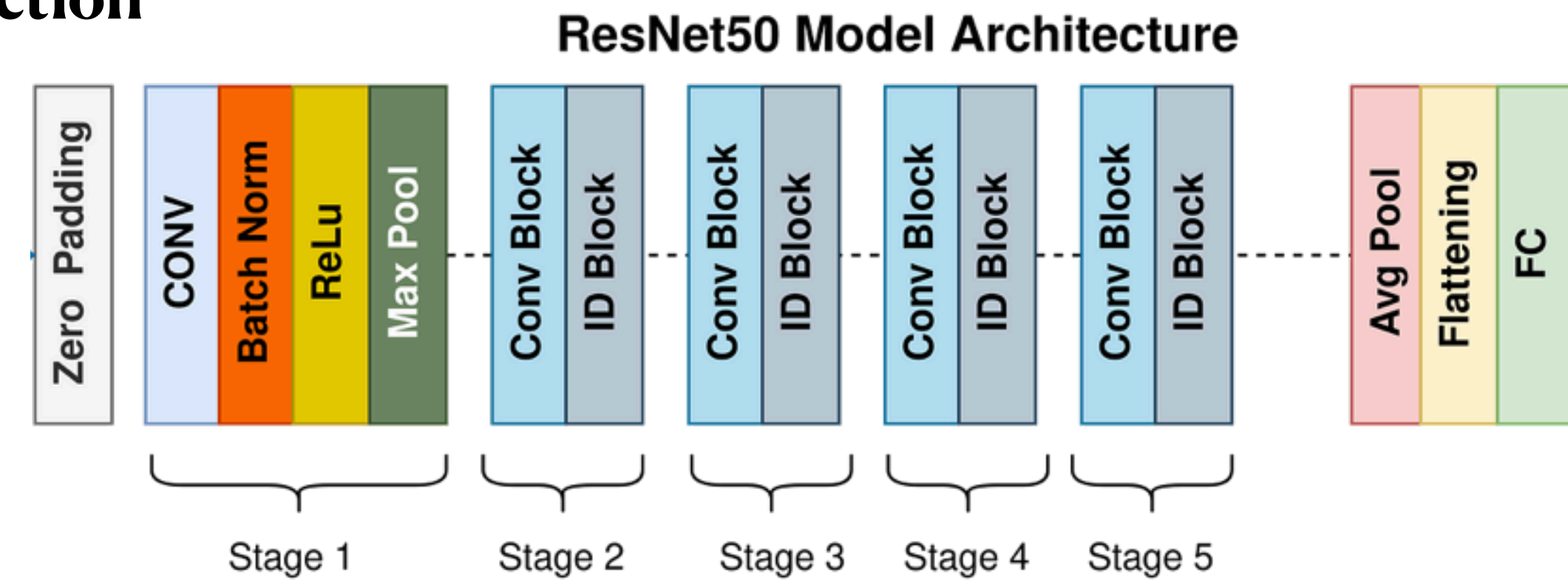
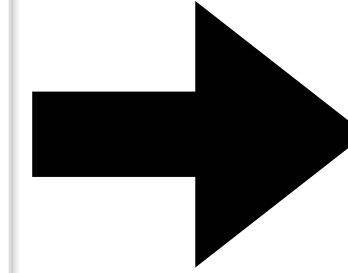
Visual Content



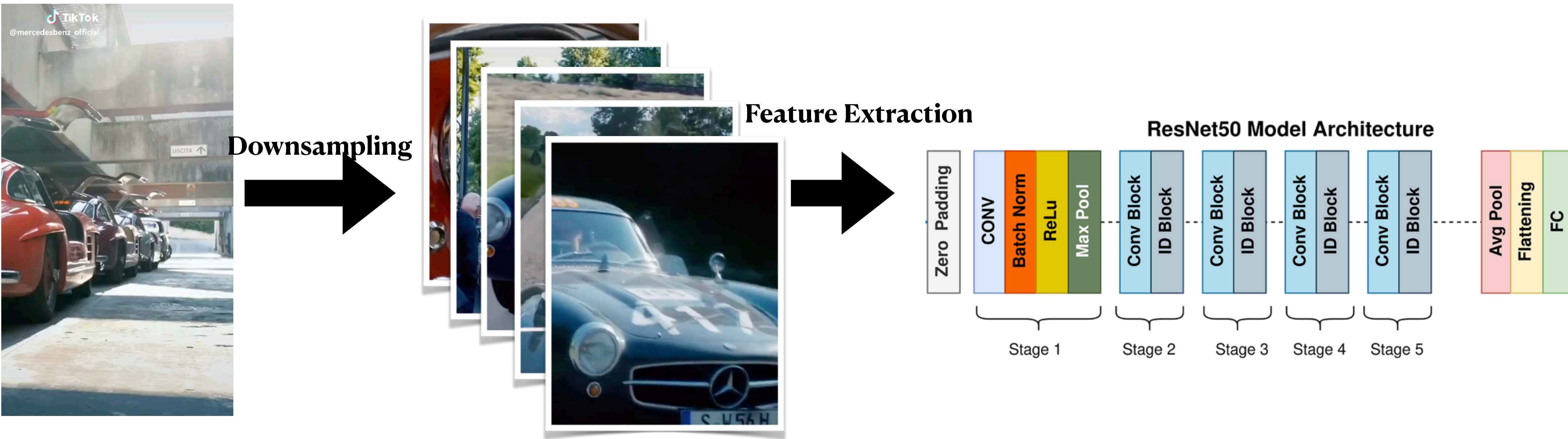
Downsampling



Feature Extraction

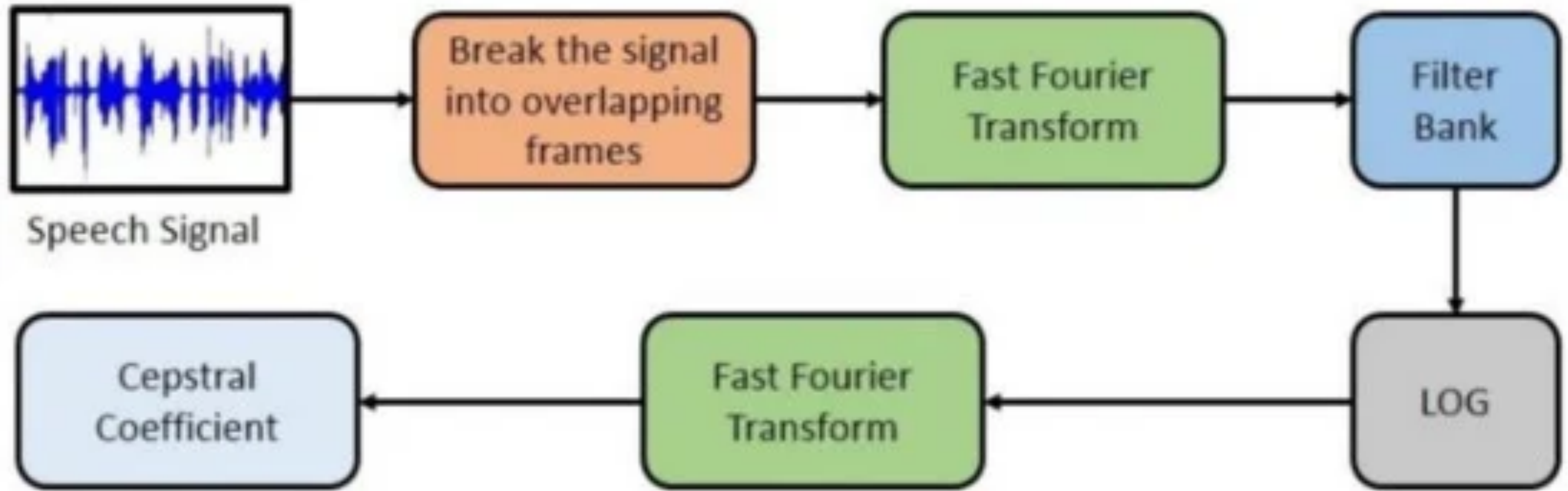


Visual Content



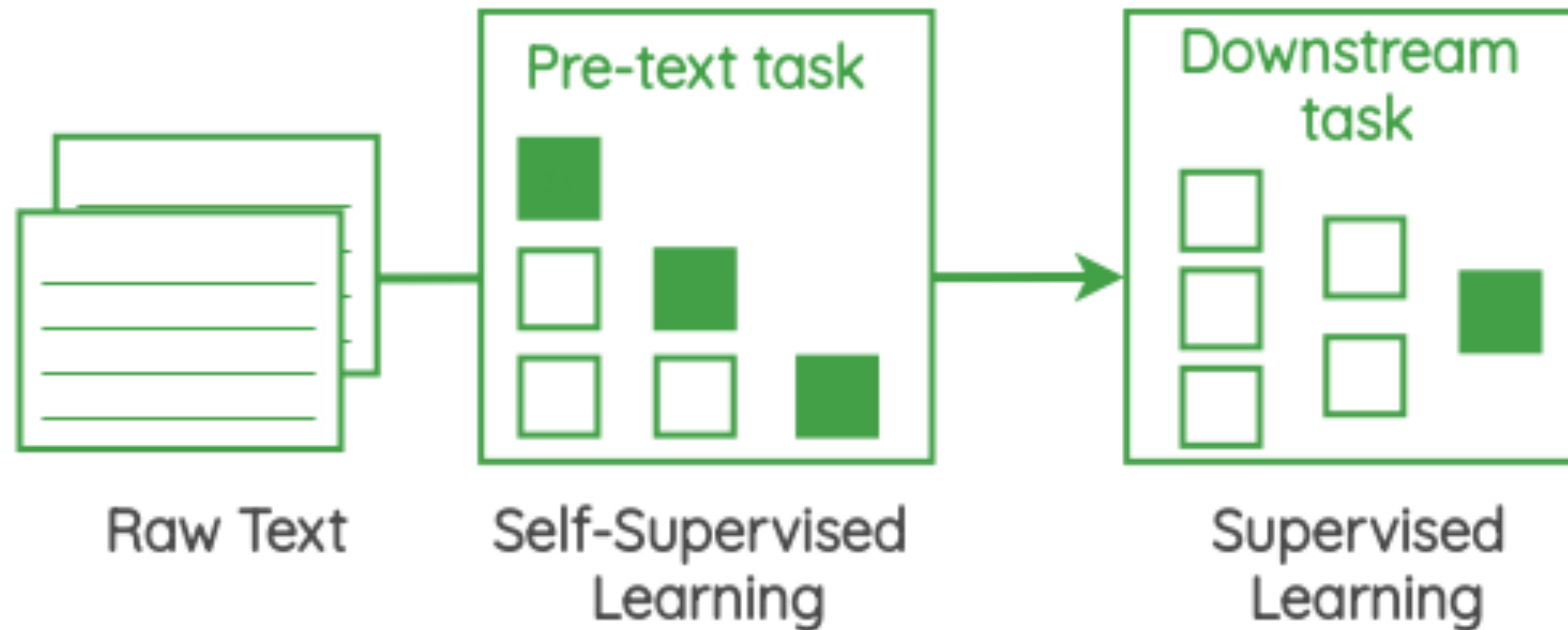
1. Perform **downsampling** to obtain **10** pictures in uniform timeframe
2. **Feature extraction** using pre-trained ResNet50 Neural Network
3. Output dimension: **512*10**

Acoustic Content



- **Extract 48-dimensional acoustic feature using Mel-Frequency Cepstral Coefficients (MFCC)**

Textual Content



- Collect all text descriptions as training dataset
- Apply contrastive learning as a black box to extract 20-dimensional features

Social Content

	Publisher ID	Publisher Follower Count	Publisher Following Count	Publisher Total Heart Count	Publisher Total Video Count	Video Create Time
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2	A11	22.3M	413	363M	1826	Feb-15-2022

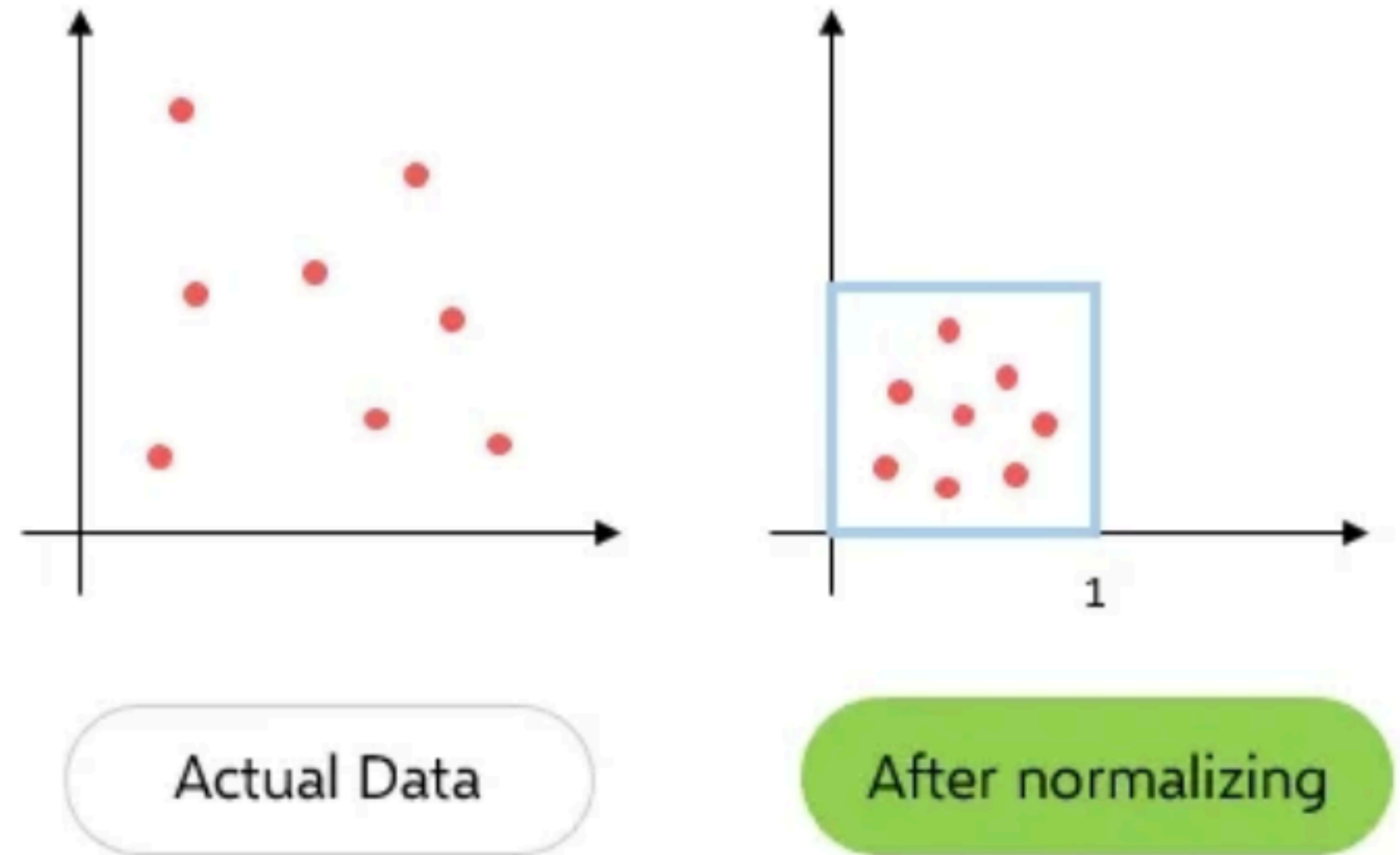
- Categorical-valued: perform one-hot encoding
- Replace “Video Create Time” by “Length of time until collected time (Mach 31, 2022)”
- Number of features: $15+5=20$

Summary

Pre-processed feature dimension:

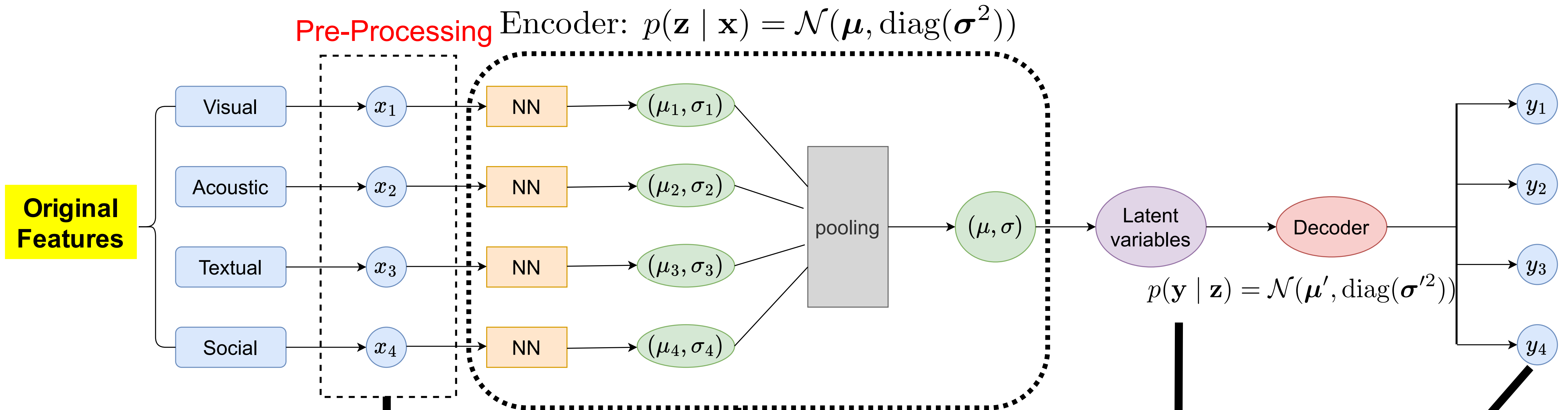
- Visual Content: 5120
- Acoustic Content: 48
- Textual Content: 20
- Social Content: 20

Normalize continuous-valued features to have zero mean and unit variance



3. Model Formulation and Methodology

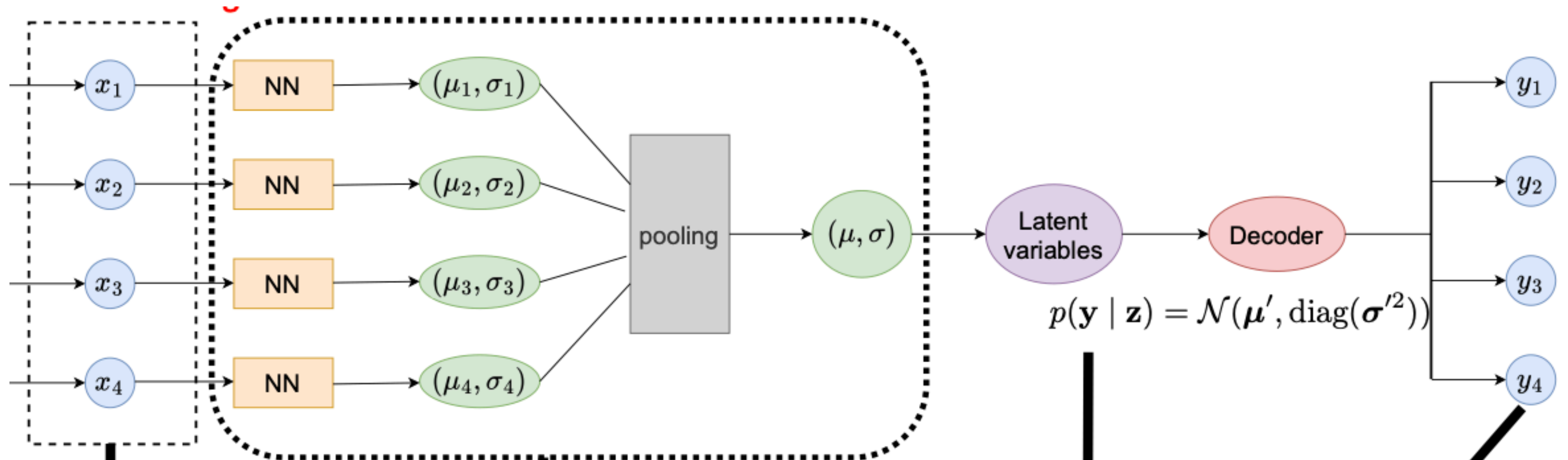
Variational Autoencoder



Variational Autoencoder

- $\mathbf{x}$: pre-processed feature
- $\mathbf{z}$: latent variable
- $\mathbf{y}$: 4-dimensional label

1. Encoder: return $\mathbf{x}$ to $\mathbf{z}$ by building $P(\mathbf{z} | \mathbf{x})$
2. Decoder: convert a randomly sampled $\mathbf{z}$ to $\mathbf{y}$ using $P(\mathbf{z} | \mathbf{y})$



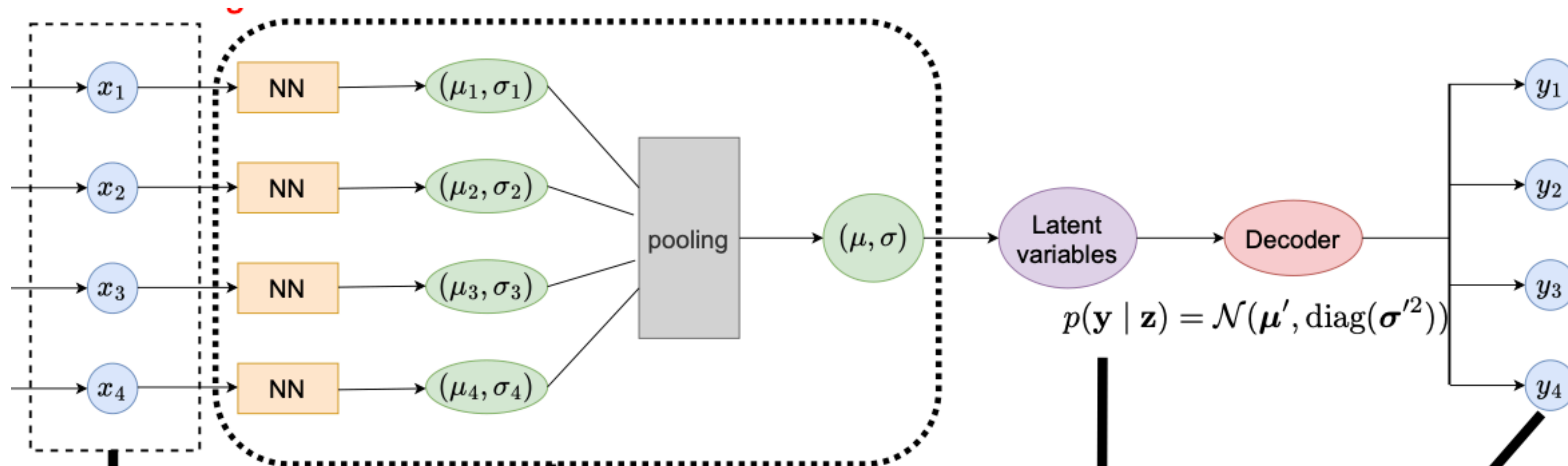
Encoder Parameterization

- Model assumption: conditional independence of features from

distinct parts:
$$p_{\theta}(\mathbf{z} \mid \mathbf{x}) \propto p_0(\mathbf{z}) \prod_{j=1}^4 p_{\theta}(\mathbf{z} \mid x_j)$$

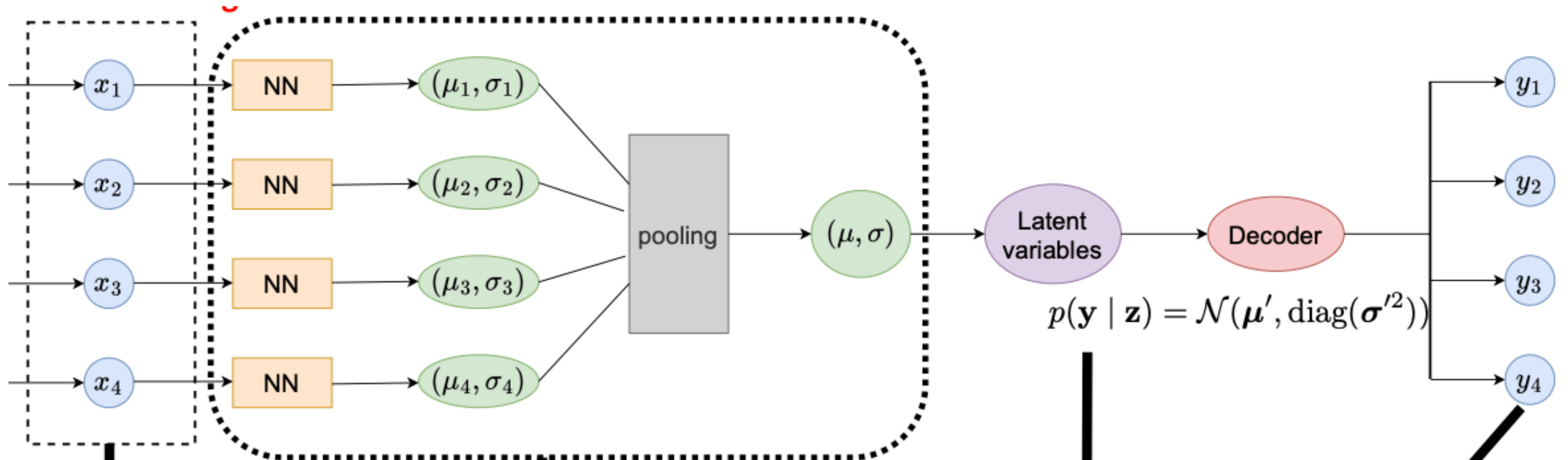
$$p_{\theta}(\mathbf{z} \mid \mathbf{x}_j) = \mathcal{N}(\mu_j, \text{diag}(\sigma_j^2)), \quad (\mu_j, \log \sigma_j) = \text{Encoder-Neural-Net}(\mathbf{x}; \theta)$$

$$p_{\theta}(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\mu, \text{diag}(\sigma^2)), \quad (\mu, \sigma) = \text{pooling}\left(\{\mu_i, \sigma_i\}_{i=1}^4\right)$$



Decoder Parameterization

$$p_{\phi}(\mathbf{y} \mid \mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}', \text{diag}(\boldsymbol{\sigma}'^2)), \quad (\boldsymbol{\mu}', \log \boldsymbol{\sigma}') = \text{Decoder-Neural-Net}(\mathbf{z}; \phi)$$



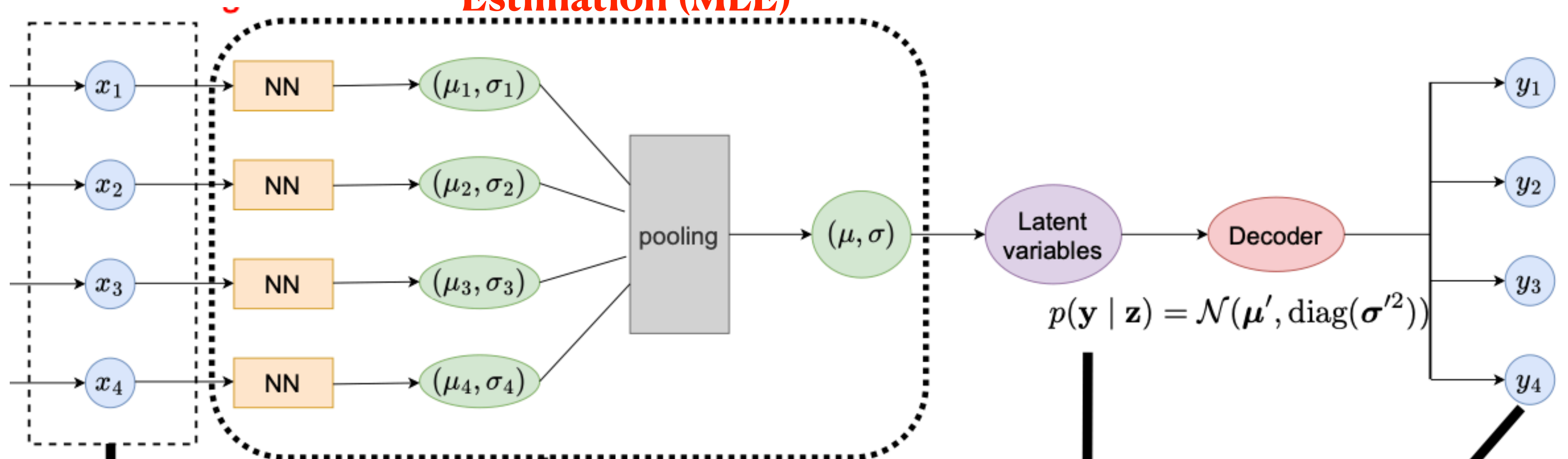
Optimization Objective

$$\max_{\theta, \phi} \left\{ \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\log p_{\theta, \phi}(\mathbf{x}, \mathbf{y}) \right] - \lambda \text{KL}(p_{\theta}(\mathbf{z} \mid \mathbf{x}) \parallel p_0(\mathbf{z})) \right\}$$

Objective is
intractable to
compute

**Maximum Likelihood
Estimation (MLE)**

Regularization

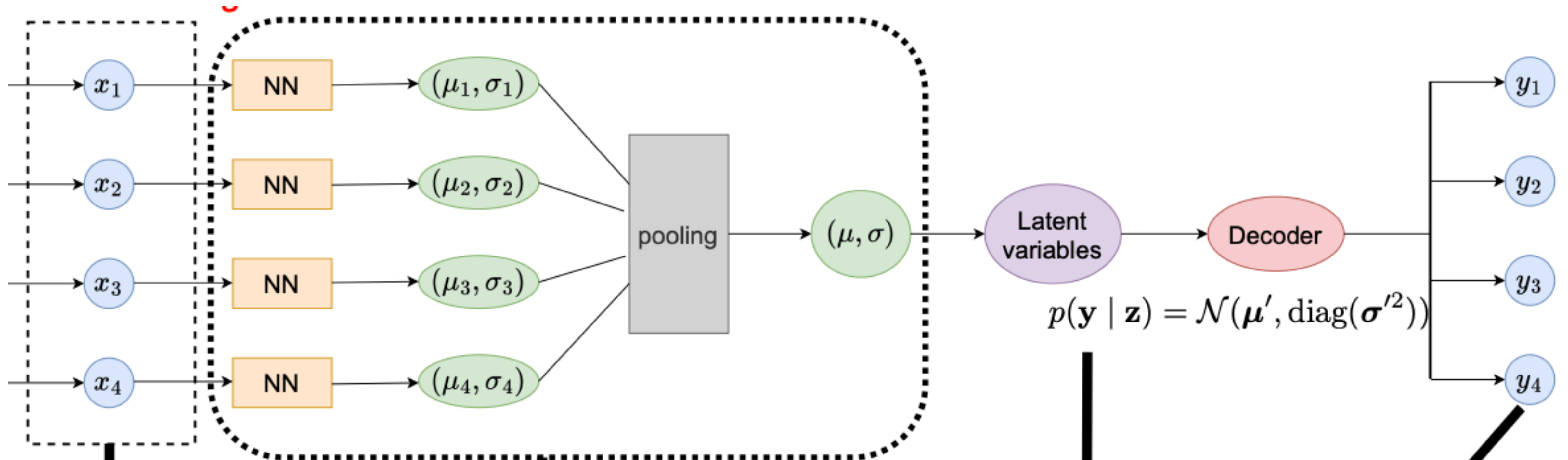


Efficient Optimization Objective

$$\max_{\theta, \phi} \left\{ \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\mathbb{E}_{p_{\theta}(\mathbf{z} | \mathbf{x})} [\log p_{\phi}(\mathbf{y} | \mathbf{x})] - \lambda \text{KL}(p_{\theta}(\mathbf{z} | \mathbf{x}) \| p_0(\mathbf{z})) \right] \right\}$$

Evidence Lower Bound of MLE

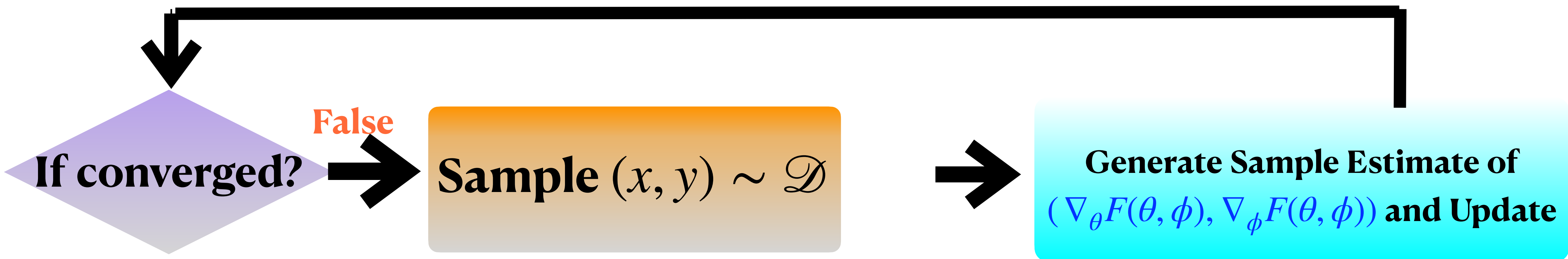
Regularization



Optimization Algorithm

$$\max_{\theta, \phi} \left\{ F(\theta, \phi) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\mathbb{E}_{p_{\theta}(\mathbf{z} | \mathbf{x})} [\log p_{\phi}(\mathbf{y} | \mathbf{x})] - \lambda \text{KL}(p_{\theta}(\mathbf{z} | \mathbf{x}) \| p_0(\mathbf{z})) \right] \right\}$$

Stochastic Gradient Method:



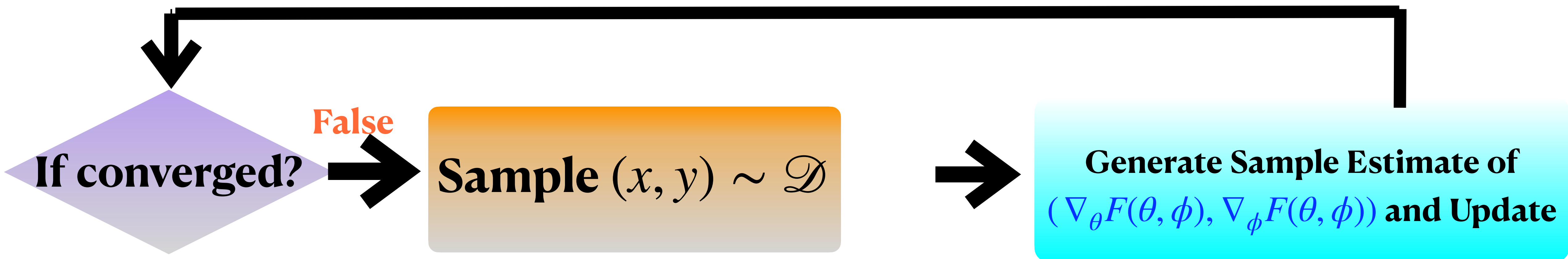
- Sample estimate of $\nabla_{\phi} F(\theta, \phi)$ is **easy**:

$$\nabla_{\phi} \mathbb{E}_{(x, y) \sim \mathcal{D}} \left[\mathbb{E}_{p_{\theta}(\mathbf{z} | \mathbf{x})} [\log p_{\phi}(\mathbf{y} | \mathbf{z})] \right] = \mathbb{E}_{(x, y) \sim \mathcal{D}} \left[\mathbb{E}_{p_{\theta}(\mathbf{z} | \mathbf{x})} [\nabla_{\phi} \log p_{\phi}(\mathbf{y} | \mathbf{z})] \right]$$

Optimization Algorithm

$$\max_{\theta, \phi} \left\{ F(\theta, \phi) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\mathbb{E}_{p_{\theta}(\mathbf{z} | \mathbf{x})} [\log p_{\phi}(\mathbf{y} | \mathbf{x})] - \lambda \text{KL}(p_{\theta}(\mathbf{z} | \mathbf{x}) \| p_0(\mathbf{z})) \right] \right\}$$

Stochastic Gradient Method:



- Sample estimate of $\nabla_{\phi} F(\theta, \phi)$ is **easy**
- Sample estimate of $\nabla_{\theta} F(\theta, \phi)$ is **difficult**: $\nabla_{\theta} \left[\mathbb{E}_{p_{\theta}(\mathbf{z} | \mathbf{x})} [\cdot] \right] \neq \mathbb{E}_{\nabla_{\theta} p_{\theta}(\mathbf{z} | \mathbf{x})} [\cdot]$

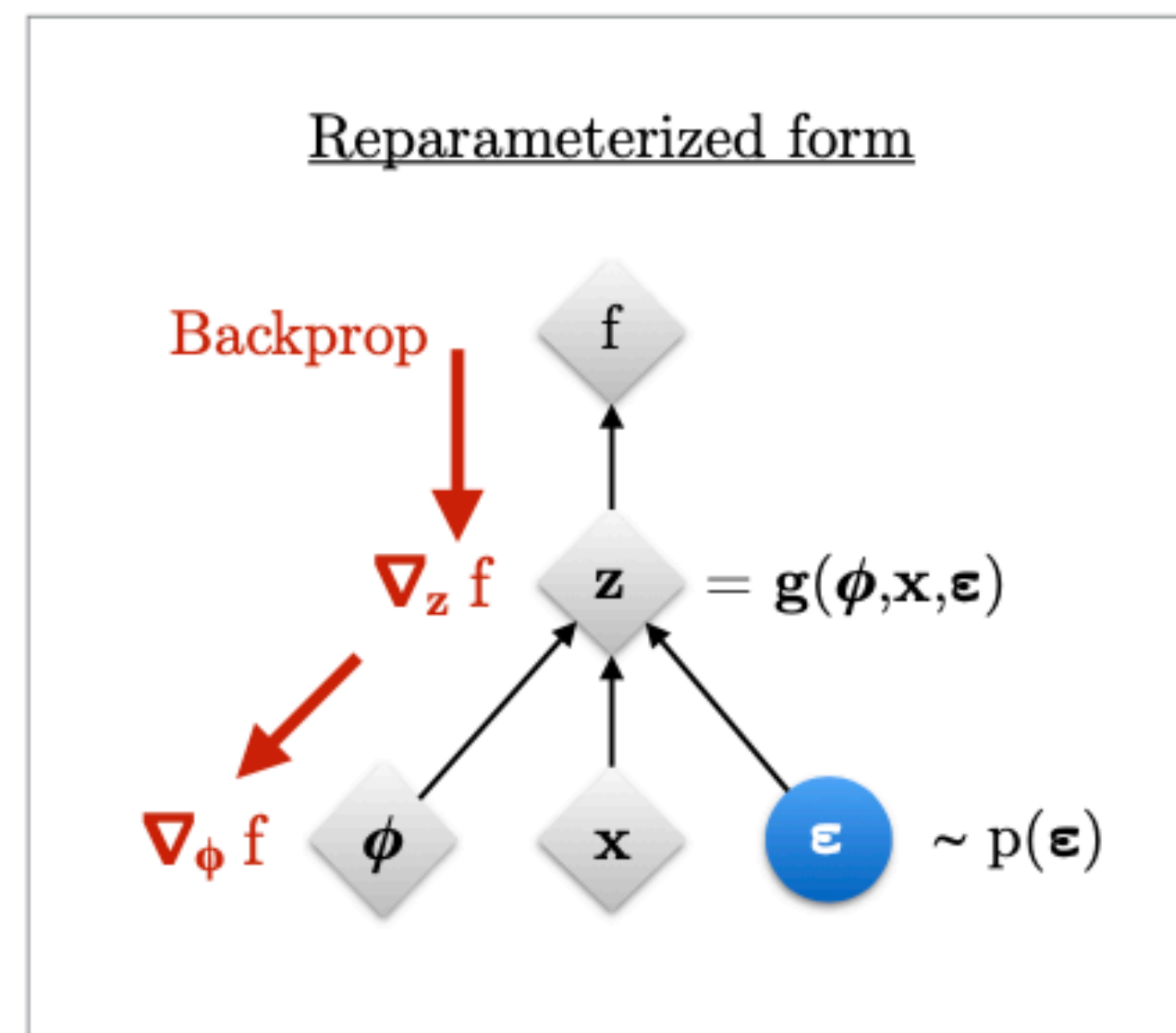
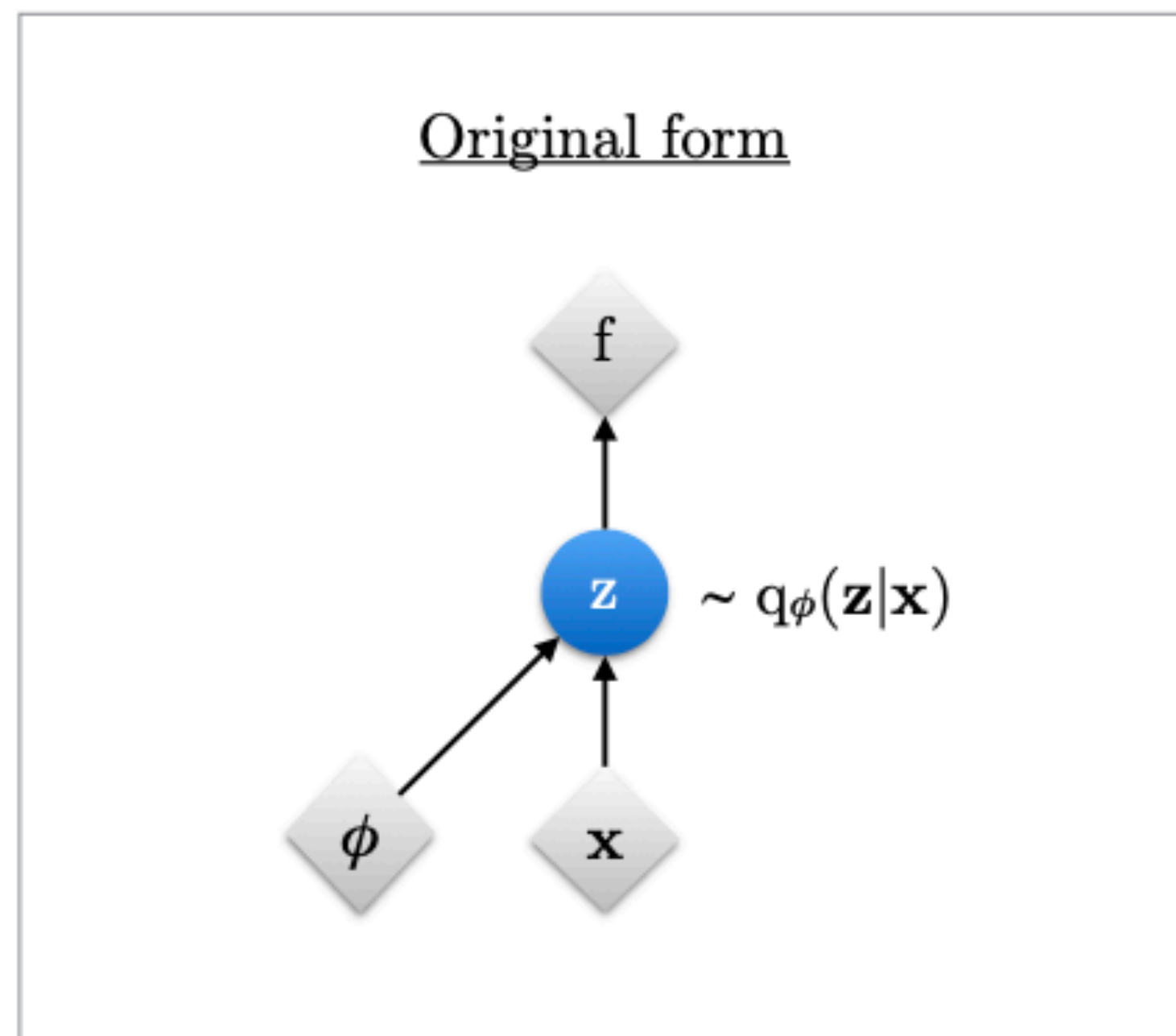
Re-parameterization Trick

- $\mathbf{z} \sim p_{\theta}(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$
- $\mathbf{z} = g(\epsilon, \theta, \mathbf{x}) := \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$

$$\begin{aligned} \nabla_{\theta} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\mathbb{E}_{p_{\theta}(\mathbf{z} \mid \mathbf{x})} [\log p_{\phi}(\mathbf{y} \mid \mathbf{z})] \right] &= \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\nabla_{\theta} \mathbb{E}_{p_{\theta}(\mathbf{z} \mid \mathbf{x})} [\log p_{\phi}(\mathbf{y} \mid \mathbf{z})] \right] \\ &= \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\nabla_{\theta} \mathbb{E}_{\epsilon} [\log p_{\phi}(\mathbf{y} \mid g(\epsilon, \theta, \mathbf{x}))] \right] = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\mathbb{E}_{\epsilon} [\nabla_{\theta} \log p_{\phi}(\mathbf{y} \mid g(\epsilon, \theta, \mathbf{x}))] \right]. \end{aligned}$$

Re-parameterization Trick

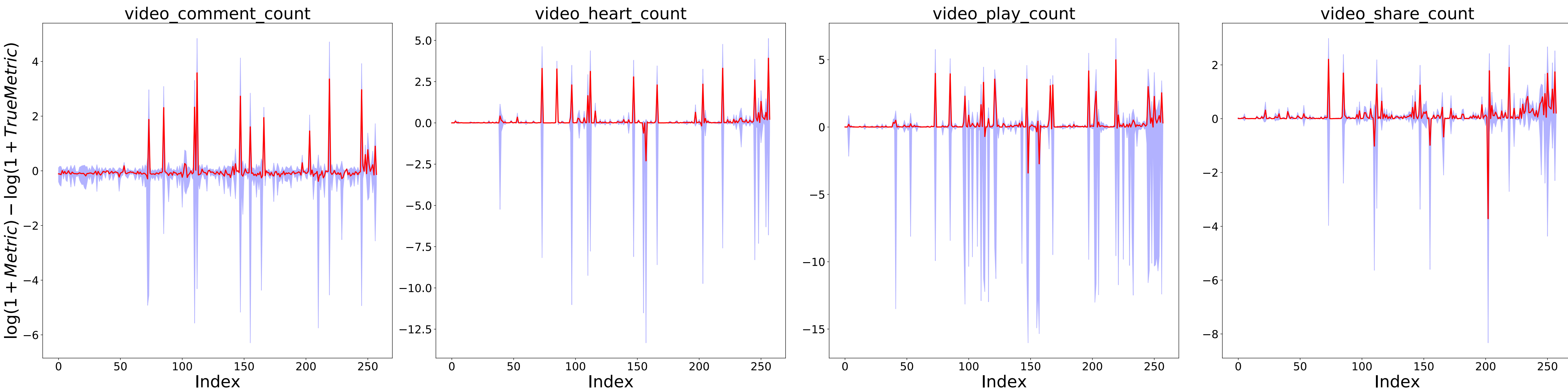
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- $\mathbf{z} = g(\epsilon, \theta, \mathbf{x}) := \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$



4. Numerical Study and Conclusion

Numerical Results

- Point and interval estimates for the popularity metrics of each video in the testing dataset
- y-axis: $\log(1 + \text{Predicted Label}) - \log(1 + \text{True Label})$



Numerical Results

- Mean Absolute Percentage Error:

$$\text{MAPE} = 100 \times \frac{1}{n} \sum_{i=1}^n \left| \frac{\text{Label}_i - \widehat{\text{Label}}_i}{\text{Label}_i} \right|$$

Video Comment Count	Video Heart Count	Video Play Count	Video Share Count
0.627	0.883	2.171	0.296

Numerical Results

Week 1 Ranking

Team	Rank
9	1
8	2

Week 2 Ranking

Team	Rank
9	1
8	2
19	3

Week 3 Ranking

Team	Rank
9	1
25	2
8	3
22	4
19	5

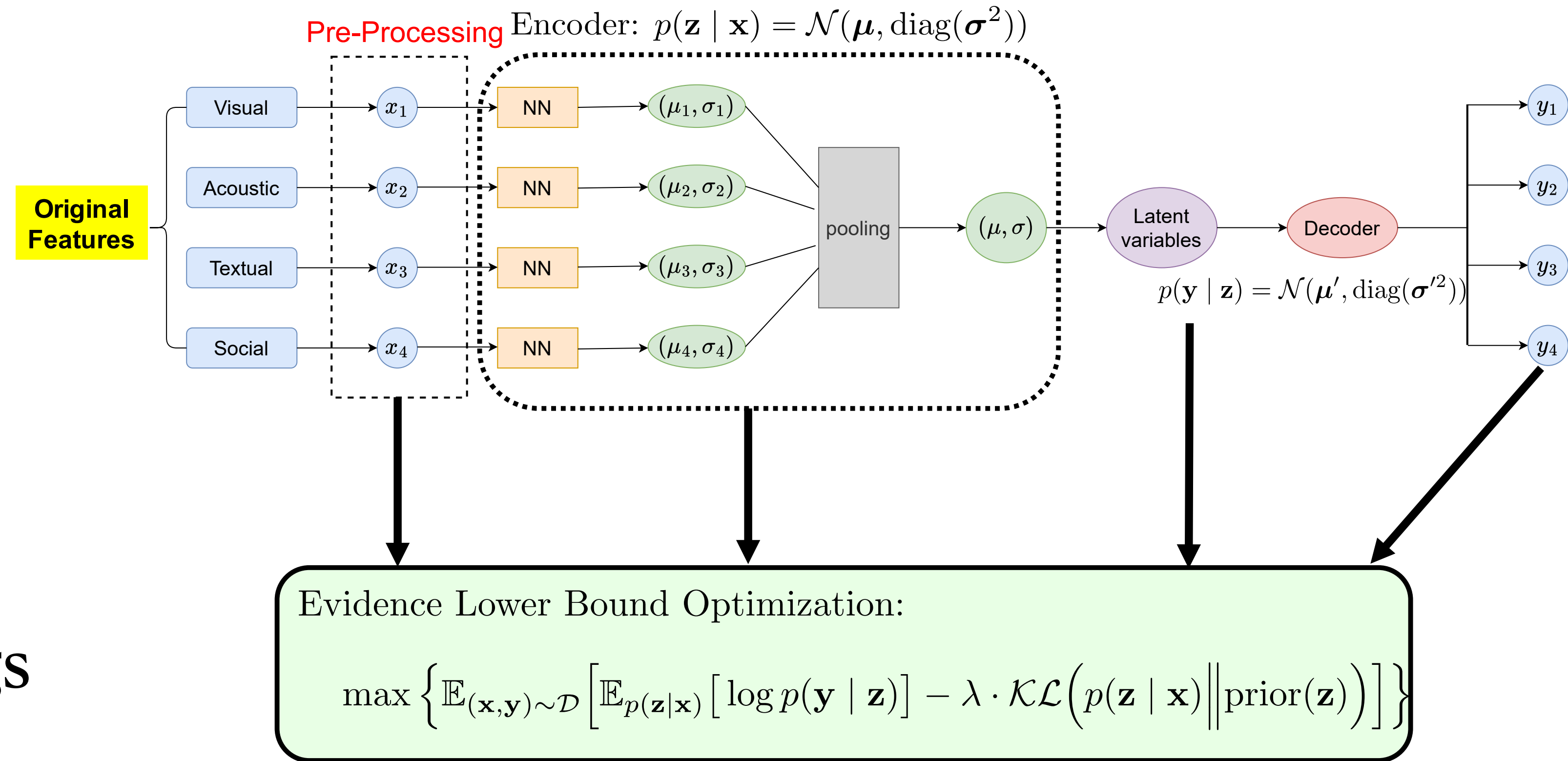
Final Ranking

Team	Rank
25	1
9	2
8	3
22	4
19	5
5	6

9 refers to our team

Conclusion

- Deep-learning based approach
- Variational inference
- Uncertainty quantification
- Second-highest rank, as validated by official rankings



Future Plans

- More flexible modeling of encoder and decoder
- Advanced structure of generative modeling

